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**Project @ Air India Hangar at Delhi**



# Quotation

ASSA ABLOY

## ASSA ABLOY Entrance Systems

The global leader in
door opening solutions

Quotation reference: **SQT/19-20/1643**

Date: 20th May 2019

**Air India**
Delhi

**Kind Attn: Mr. Sanjay Sharma**
**Contact No:**
**Email:** Sanjay.sharma1@airindia.in

Project: Air India Hangar at Delhi

Dear Sir,

Thank you for giving us an opportunity to participate with our offer on the above project. We enclose herewith our proposal which includes:

A. Technical Proposal

By choosing ASSA ABLOY Entrance Systems as the supplier of your door automation products, you will receive the following advantages:

- You are working with a Global Provider with financial strength and over 50 years' experience of the global market place in Entrance Solutions.
- World leading products ensure excellent reliability and life-cycle costs to give you peace of mind.
- A comprehensive product helping to provide value engineered solutions

**Major Benefits using Megadoors**

- No floor track (and therefore no concrete grade beam) needed; a substantial saving in civil cost.
- Faster opening and closing time due to vertical lifting, resulting in a substantial reduction in loss of energy.
- Minimal side-room required (no "door pockets"); a substantial saving in space and cost of land, this is especially relevant, when a series of hangars are built adjacent to each other.
- The doors have very tight sealing, to protect against water, sand and dust ingress, which is essential for a maintenance hangar.

## Technical Specifications :

### Hangar purpose, type of aircraft

The hangar will be used for the storage of Aircraft. The hangar door must provide a tight seal to reduce energy loss and provide protection against ingress of dust, sand and water. A short opening and closing time of the door will contribute to the saving of energy.

### Opening dimensions

Clear opening required for Hangar Door: **(W) 40m * (H) 13.5/10 m – Qty 1 No**

### Space limitations

The hangar has with very limited space to the sides of the hangar, therefore a vertically lifting door, is suitable for the project.

### Wind load

The doors will be designed to take into account the local environmental conditions i.e. high winds, high temperatures, high levels of dust and high humidity levels.

- ☐ ***Door sections designed for a wind load of 1kPa***

### Quality Standards

The hangar doors will contribute to a very clean working environment. Our hangar doors have a proven track record of similar installations and provide quality certifications on request.

### Other vehicle traffic

Although the hangar doors are designed to accommodate the size of the specified aircraft, part of the doors must be opened for other vehicle traffic; the opened part must be reduced to a minimum level.

### Door construction

The doors will be constructed from materials, which resist corrosion so as to minimize long term maintenance costs.

**B. PROPOSAL:**

Supply of Aircraft Hangar Doors from NIHVA ASSA ABLOY Entrance Systems, offering Megadoor as product brand.

To suit the door opening requirement, we propose Megadoor VL 3190 vertical lifting hangar door with the following configuration.

**Hangar Door: ASSA ABLOY VL3190 Megadoor with 3 sections and 2 swing up mullion**

| Door | Door Leaves | Clear opening Width (M) | Clear Opening Height (M) |
|---|---|---|---|
| Door 1 | 3 | Total :40 | Max :13.5/10 |

The dividing of the door sections and the sizes may vary according to the final approved design of the building.

**Wind load:**

☐ ***Door sections are designed for a wind load of 1kPa***

**Short specification:**

A. **General**: Vertical Lifting Fabric Doors door are designed in such a way that the material, construction, and operation of the door confirm to the performance specifications contained herein and it doesn't consist of any protruding parts affecting personal safety in site. Doors operate properly without binding, interference, or damage to the adjacent structure.

B. **Door Leaf:** The door leaf is made of two layers of very strongly vinyl coated polyester fabric weighing approximately 700g/sq.m), capable of carrying 2400N/5cm. It is UV stabilised, self extinguishing (0 -25 flame spread), and suitable to withstand temperatures between – -30°C to +70°C. The translucent material if used meeting the same requirements and loading as the standard door fabric. The fabric attached to both sides of the intermediate beams, top beam and bottom beam with self tapping screws through corrosion resistant aluminium batten strips Powder coated clamp strips to match the color of the fabric. The door fabric is pulled tight between the intermediate beams when the door is fully closed and wind lock engaged. The fabric is tested to meet the standards as below :

- **Flame retardency:** tested according: EN 13501-1, B-s2,d0.
- **Tensile strength:** 2500/2000 N / 5 cm acc to EN ISO 1421-1
- **Tear resistance:** 400 / 300 N acc to DIN 53363
- **Light resistance:** ≥ 7 acc to ISO 105-B02
- **Radar reflecting coefficient:** 0,3dB, ~0,1%

C. **Door Color:** Our standard colors are - Red, Blue, Grey, White, White aluminum, Tan, Green, Graphite or Translucent.

D. **Doors:** It consist of hoist up fabric doors with intermediate aluminium beams or trusses. Fabric is gathered above the head of the opening. Regardless of location, the minimum design wind load is calculated to 300km/hr.

E. **Door Beams:** The intermediate beams are corrosion resistant extruded aluminium tempered after the extrusion process to ensure required strength and have a length to depth ratio not to exceed 100:1 to meet the required wind load, calculated according EN 13241-1 with beam safety factor 1.5 X EN 12604. Beams are spaced 700 to 1400 metres apart, dependent upon the wind load. At each end of the beams there is a guide block of self-lubricating nylon material, or rollers that run along the guides. The intermediate beams must have a minimum depth of 290mm. The intermediate door beams and guide block or rollers are designed to carry the full design wind load without failure or being pulled from the door guides. There are no multiple lifting cables and holes distributed along the length of the intermediate beam.

F. **Bottom Section:** The bottom section is made of steel. A rubber seal will be fixed to the section to ensure tightness against the floor /ground. Bottom section should prevent lifting of door during wind load and secured at locked position with an automatic wind lock system.

G. **Door Guide:** The vertical guides are an integral part of the door, made of extruded aluminium with a suitable depth and width dependent on the size of the intermediate beams. The guides is designed to provide weather sealing on the inside and the outside faces. The guides is designed to provide sufficient strength to transfer the wind load and safety arrestor forces from the door beams, at maximum wind load, to the jamb steel support system. The guides is easily replaceable in case of damage.

H. **Drive Units:** The door leaf have an IEE / EN labelled single or dual motor drive system with torque sized as appropriate for the weight of the door leaf. The gear motor is equipped with a drum on which the belts/ropes are wound. If a single motor drive is used, the belts/ropes is wound on the same drum. The belts/ropes is attached to the bottom beam via the safety arrestors. The gear motor is removable without disturbing the limit switch adjustment.

I. **Belt System:** A maximum of two belts/ropes per door, running inside the doors guides is used to transmit motive force to the door unit. Similarly, the quantity of sheaves used to guide the belts/ropes is minimised to reduce maintenance requirements and spare parts inventories.

J. **Safety Arrestors:** Each door leaf will be supplied with two safety arrestors that activate and support the door in the event a belt/rope should break. Each safety arrestor type is tested to the standards established by TUV-control for the required door leaf weight and hold a valid certificate. The movable mullions, when provided, will be provided with a lifting motor that includes back up arresting system to prevent the mullion from falling in case of chain or rope rupture.

K. **Slack belt Breaker:** A safety device is used on all door leaf belts/ropes which will sense a slack belt/rope condition and cut power to the appropriate drive unit to prevent an unsafe condition.

L. **Bottom Beam:** The bottom beam is designed with a suitable width and depth to carry the load of the intermediate beams when the door is open, and to ensure full closing and a tight floor seal in heavy winds. A heavy U-shaped bottom rubber will be provided to form a tight seal with the floor, even on uneven surfaces.

M. **Self Activating Wind Lock:** Each door leaf will be provided with two self-activating wind locks which activate and lock the bottom beam when the door reaches its closed position. This locking action will act to maintain a tight floor seal and intermediate stability even under conditions of heavy wind. A proximity switch, with indicator is provided to notify the door operator that all wind locks are engaged properly.

N. **Header Box:** Header boxes, when provided, will be constructed of carbon steel and factory painted to C3 environmental standards with rust inhibitive paint.

O. **Control Panel:** Our control panel contain interlocks to preclude personnel injury, including interlock between the power supply system and use of the hand crank for manual operation of the door unit. The control panel is mounted a minimum of 1200mm above the floor. The control panel will be IP 65 rated. The control panel will be a PLC based system.

P. **Swing up Mullion:** The swing-up mullions will be designed to swing up under the door leaf in the raised position. They are constructed of steel and painted to fulfil environmental Class C3 according to ISO 12944-2. The mullion hinge pivot is of bearing design. The mullion hinge pivot will be of a maintenance free bearing design. The electrical control panel will provide an interlock function to coordinate door leaf and mullion operation to prevent equipment damage. Mullion push buttons for both up and down operation will be of the constant pressure type. The mullion will open/close automatically directly from the control panel. The mullion will not require any manually locking in closed position.

Q. **Wind Resistance:** Door will be engineered to take into account the local environmental conditions i.e. high winds, high temperatures, high levels of dust and high humidity levels.

R. **Water penetration:** Door will be engineered in accordance to EN12425 Class 3 with resistance to water penetration of 110PA for a closed door with or without pass door. The door will be exceptionally designed to withstand water spray for >20minutes under pressure of >50PA ($N/m^2$).

S. **Air Tightness:** Doors will be engineered in accordance to EN 12426 **Class 2- 3,** with or without pass-door.

T. **Door Operations**

**1. Motor**

a. Doors will be provided with an electrically operated motor and controls wired for operation on voltage indicated on the electrical drawings.

b. Motors will be heavy duty industrial type with IP55 protection class.

c. The control panel will have an optional facility to provide generator backup.

d. Motor Power Rating: Motor will be of the manufacturer's standard power rating.

e. The power supply will be 03 phase with 400V/50Hz input supply**.**

**2. Door Operation:** The vertical lifting fabric door will guide up and down in the weather sealing vertical guides attached to the structure. The door will operate by lifting the bottom beam upwards, thereby stacking the intermediate beams one on top of the other, with the fabric panel folding in pleats. The fabric panel will go over the top beam, covering both sides of the door, and will be attached to the intermediate beams by screws and batten strips. When the door is fully closed, the intermediate beams will hang between the two fabric door panels thus pulling the fabric tight. The tension created in the fabric panels will stabilize the intermediate beams. The safety arrestors will be attached to each end of the bottom beam and will travel in the vertical guide tracks. The safety arrestors will immediately stop the downward movement of the door in case of belt/cable failure.

**3. Electrical Operation:** Audible and visual alarm will activate five seconds prior to door movement and during door movement. When the door is completely opened it will stop on the primary top limit switch. In case of over travel, a secondary limit switch will cut off the drive unit's motor current to prevent damage. These two limit switches will be activated by the uppermost intermediate beam. The drive unit will be stopped by the slack belt/cable breakers when the door is closed. The slack breakers will also stop the door in case of belt/cable rupture or if an obstruction should prevent the door from being closed. The slack breakers will be activated by weight or springs. The door and mullion operation will be controlled by a 10-12" LCD touch screen and by four buttons marked "Open", "Close", "Stop" and E-Stop. All doors and mullions will be operated by first selecting the component via the LCD touch screen then pressing the open, close or stop button. The doors will be controlled by momentary pressure to open and constant pressure to close. Removing the pressure from the close button will stop the motor drive and set the brake. The mullion operation will be controlled by constant pressure to open and to close. Removing the pressure from either button will stop the motor drive and set the brake. The door manufacturer will provide an enclosure which is factory wired and equipped with instantaneous overload relays.

4. **Auxiliary Emergency Operation**: The control panel will be fed from a transfer switch that will supply emergency power to the door systems in case of a power failure. The control panel will have a set of 230V terminals which can be connected to a dry contact located in the customer's transfer switch. The contact will be open when the transfer switch is in the normal and center position. The contact will close when the transfer switch is in the emergency power position. When in the emergency power position door operation will be one-door-at-a-time operation to prevent emergency power overload. The wiring from the control panel to the automatic transfer switch will be by the Electrical Section.

5. **Controller**

   a. Doors will be supplied with PLC based control unit. The control unit will commands the gear motor via push buttons. The UP button opens the doors by impulse signal and DOWN button is set to hold to run, rated with protection class IP 65.

   b. Fire alarm integration: Overhead hoist up fabric doors will be provided with an additional provision of FA integration. Automatic opening of door in case the fire alarm give a signal.

   c. Diagnostic program that will allow the controller to identify potential breakdown continuously 24/7 to identify problem and allow for changes before it brakes.

   d. The controller will allow for SIM -card or Ethernet connection so that the local service team gets online reporting from the door.

6. The drive unit will be connected to the owner furnished and owner installed emergency generator,

7. The control panel will be wired to accept a furnished plug-in generator to operate doors in case of a power failure. (The operation of only one door at a time is required.)

U. **Finish**

1. Steel components will be Class C3 accordance to ISO 12944-2
2. Door leaf screw to be corrosion protected with Geomet.
3. Concealed ferrous metal surfaces will be hot dip galvanized.

If you have any questions after reading this proposal, or if you need more information, please do not hesitate to contact **Mr. Vijay Tarak +918551093333 / Mr. Ashish Nair +971506320495**

Kind Regards,
**Ms. Varsha**
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